
TN2: THE TWO-QUBIT SINGLET CORRELATION AS A SYMMETRY-AND-OPERATIONS SPECIALISATION OF DETERMINISTIC VOLUME GEOMETRY

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Zayn Blore

 zayn.blore@ou.ac.uk 
www.constraintsurfacedynamics.com

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ABSTRACT

We study the two-qubit singlet state as a finite-dimensional specialisation of the symmetry-and-operations route to quantum probabilities in deterministic volume geometry. The purpose of the present note is limited and explicit: to verify that the singlet preparation in dimension 4 satisfies the operational hypothesis package relevant to the finite-dimensional Born-weight selection layer, and hence that the admissible probability assignment takes the Born form in this case.

Using that Born-form assignment for the singlet preparation, we evaluate the joint outcome probabilities for arbitrary detector directions $a, b \in S^2$ and obtain

$$E(a, b) = -a \cdot b.$$

The standard CHSH value $2\sqrt{2}$ then follows for the usual coplanar choice of settings.

The logical division of labour is explicit. The repeated-trial typicality bridge from invariant weights to limiting frequencies belongs to Paper A and LF1. The measure-bridge and finite-dimensional Born-weight layer belong to TN1 and Paper B. The constructive system-apparatus Hamiltonian model for the singlet is deferred to TN3, and the Bell-premise diagnosis is deferred to c1. The present note should therefore be read as the non-constructive singlet specialisation of the finite-dimensional symmetry-and-operations route, not as a constructive measurement model.

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1 Introduction

The two-qubit singlet state is the natural finite-dimensional composite test case for the symmetry-and-operations probability route developed elsewhere in deterministic volume geometry. The purpose of the present note is narrow. We do not construct a system-apparatus Hamiltonian here, and we do not develop the Bell-premise diagnosis in full. Instead, we verify that the singlet setting in dimension 4 falls within the scope of the existing finite-dimensional measure-and-weight layer, and we derive the resulting joint probabilities and correlation function.

The logical division of labour should be kept explicit. The repeated-trial frequency bridge from invariant ontic weights to operational frequencies belongs to Paper A and LF1. The measure bridge and the finite-dimensional Born-weight selection layer belong to TN1 and Paper B. The constructive system-apparatus singlet model is deferred to TN3. The Fine-style Bell diagnosis for contextual outcome maps is deferred to c1. The present note should therefore be read as the prior non-constructive singlet specialisation on which those later analyses build.

The load-bearing claim is the following. For the singlet preparation in $H_{AB} \cong \mathbb{C}^4$, the induced operational probability assignment satisfies the singlet-specialised structural package relevant to the existing finite-dimensional probability layer. It follows that the admissible probability assignment takes the Born form, and direct evaluation then yields

$$E(a, b) = -a \cdot b.$$

The standard CHSH value $2\sqrt{2}$ follows immediately for the usual coplanar settings.

The paper is organised as follows. Section 2 records the composite projective setting and the induced operational probability functional. Section 3 specialises the resulting Born-form assignment to the singlet and derives the correlation function. Section 4 records the CHSH consequence and places the result within the wider finite-dimensional programme. Appendix A provides the continuity lemma used in the covariance argument, and Appendix B verifies the singlet-specialised hypothesis package used in Section 2.5.

2 Composite Geometry and Operational Probability Functional

2.1 Composite Projective Structure

Let $\mathcal{H}_A \cong \mathbb{C}^2$ and $\mathcal{H}_B \cong \mathbb{C}^2$ denote the Hilbert spaces of the two subsystems. The composite space is

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \cong \mathbb{C}^4,$$

with associated projective manifold $\mathbb{CP}^3$.

In the underlying deterministic formulation, the composite system is represented by a joint ontic manifold Σ_{AB} equipped with Liouville measure μ_L . In the finite-dimensional quantum-effective sector, observational access is modelled by a measurable projection

$$\pi : \Sigma_{AB} \rightarrow CP^3,$$

which associates ontic microstates to projective rays. Blore [2026d,e,f] For an ontic preparation region $\Omega_{\text{prep}}^\Sigma \subset \Sigma_{AB}$, the induced epistemic preparation measure is

$$\nu_{\text{prep}}^{\text{ep}}(A) = \frac{\mu_L(\Omega_{\text{prep}}^\Sigma \cap \pi^{-1}(A))}{\mu_L(\Omega_{\text{prep}}^\Sigma)}, \quad A \subset CP^3.$$

The composite constraint surface is not a Cartesian product $\Sigma_A \times \Sigma_B$. Its symplectic structure does not decompose as a direct sum of subsystem forms. Consequently, entangled preparations correspond to non-factorisable regions of Σ_{AB} and project to non-separable regions in $\mathbb{CP}^3$.

2.2 Measurement Contexts and Projector Structure

For local measurement directions $a, b \in S^2$, define the joint eigenstates

$$|\phi_{st}(a, b)\rangle = |a_s\rangle \otimes |b_t\rangle, \quad s, t \in \{+, -\},$$

with corresponding rank-one projectors

$$\Pi_{st}(a, b) = |\phi_{st}(a, b)\rangle\langle\phi_{st}(a, b)|.$$

The four projectors form an orthogonal resolution of the identity:

$$\sum_{s,t} \Pi_{st}(a, b) = I.$$

Operationally, a measurement context $M_{a,b}$ corresponds to a macroscopic apparatus configuration. In the deterministic description, this configuration induces a measurable partition of Σ_{AB} into disjoint ontic outcome basins

$$\{\Omega_{st}^\Sigma(a, b)\}_{s,t}.$$

Under projection to CP^3 , these induce measurable epistemic outcome regions

$$\Omega_{st}^{\text{ep}}(a, b) = \pi(\Omega_{st}^\Sigma(a, b)),$$

which form a partition up to null sets of the accessible preparation support in CP^3 .

Different contexts (a, b) correspond to different partitions. No global refinement across incompatible contexts is assumed Blore [2026d,e].

2.3 Volume-Induced Probability Functional

Within the present programme, the repeated-trial typicality bridge is supplied by Paper A and formally sharpened in LF1 Blore [2026b,a]. Those results show that, under repeated preparation with invariant ontic measure, empirical frequencies converge to the corresponding normalised ontic weights. In the present finite-dimensional note, we work at the induced epistemic level and represent the corresponding operational probabilities by a context-indexed functional on the relevant outcome projectors.

Let $\Omega_{\text{prep}}^\Sigma \subset \Sigma_{AB}$ be the ontic preparation region and let

$$\nu_{\text{prep}}^{\text{ep}}(A) = \frac{\mu_L(\Omega_{\text{prep}}^\Sigma \cap \pi^{-1}(A))}{\mu_L(\Omega_{\text{prep}}^\Sigma)}, \quad A \subset CP^3,$$

denote the induced epistemic preparation measure on projective space. For a measurement context $M_{a,b}$, let

$$\Omega_{st}^{\text{ep}}(a, b) \subset CP^3$$

be the measurable epistemic outcome region corresponding to joint outcome $(s, t) \in \{+, -\}^2$. The associated operational probability is then

$$P_{st}(a, b) = \nu_{\text{prep}}^{\text{ep}}(\Omega_{st}^{\text{ep}}(a, b)).$$

Equivalently, if $\nu_{\text{prep}}^{\text{ep}} \ll \mu_{FS}$ with Radon-Nikodym density ρ_{ep} , one may write

$$P_{st}(a, b) = \int_{\Omega_{st}^{\text{ep}}(a, b)} \rho_{\text{ep}}([\psi]) d\mu_{FS}([\psi]).$$

This is the form used in the finite-dimensional operational dictionary Blore [2026d,e]. The present note does not derive this layer afresh. It applies it to the singlet setting in CP^3 .

2.4 Structural Properties

The induced operational assignment has the following properties.

(i) Finite additivity. Because the regions $\Omega_{st}^{\text{ep}}(a, b)$ form a measurable partition up to null sets,

$$\sum_{s,t} \nu_{\text{prep}}^{\text{ep}}(\Omega_{st}^{\text{ep}}(a, b)) = 1.$$

(ii) Regularity. Smooth dependence of the apparatus interaction on (a, b) induces smooth deformation of the relevant partition boundaries. Absolute continuity of $\nu_{\text{prep}}^{\text{ep}}$ with respect to μ_{FS} then gives continuity of the induced probabilities.

(iii) Residual symmetry covariance. For the singlet setting, the relevant covariance is the diagonal $SU(2)$ action associated with simultaneous rotation of both detector directions. This is the residual symmetry used later to constrain the correlation function to depend only on $a \cdot b$.

(iv) Contextual but operationally consistent assignment. Although the regions depend on the measurement context, the probability attached to a fixed macroscopic outcome channel is independent of the choice of compatible orthogonal completion. Thus the assignment is additive on each orthogonal resolution while remaining contextual across incompatible contexts.

2.5 Application of the Uniqueness Theorem

We now invoke the finite-dimensional probability-selection result of Paper B together with the measure-bridge support note TN1 Blore [2026c,f]. In that framework, the relevant structural package consists of normalisation on orthogonal resolutions of the identity, finite additivity on orthogonal projectors, Gleason-type noncontextuality within a fixed measurement context, continuity, and the relevant covariance assumptions. For finite dimension $n \geq 3$, these conditions constrain the admissible probability assignment to the Born form.

In the present note we do not reprove the general theorem Blore [2026c]. Instead, we verify in Appendix B the singlet-specialised hypothesis package required for the application in $H_{AB} \cong \mathbb{C}^4$. It follows that the induced operational probability assignment for the singlet setting takes the Born form

$$P_{st}(a, b) = \text{Tr}(\hat{\rho}_{\Psi^-} \hat{\Pi}_{st}(a, b)),$$

where

$$\hat{\rho}_{\Psi^-} = |\Psi^-\rangle\langle\Psi^-|$$

and $\hat{\Pi}_{st}(a, b)$ are the joint rank-one outcome projectors determined by the measurement context.

3 Singlet Preparation and Structural Consequences

3.1 The Singlet State on CP^3

The singlet state is

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle),$$

viewed projectively as a ray $[\Psi^-] \in CP^3$.

In the sharp-preparation idealisation, the singlet preparation is represented by the rank-one density operator

$$\hat{\rho}_{\Psi^-} = |\Psi^-\rangle\langle\Psi^-|.$$

Alternatively, for smeared or mixed preparations, one may work with an epistemic measure ν^{ep} concentrated near the singlet ray, with Radon–Nikodym density ρ_{ep} relative to μ_{FS} ; the sharp-preparation limit recovers the rank-one operator description. In either case, the resulting probabilities are state-conditioned.

3.2 Born Form of the Singlet Probabilities

From Section 2, the induced probability functional must take the Born form:

$$P(st \mid a, b, \Psi^-) = \text{Tr}(\rho_{\Psi^-} \Pi_{st}(a, b)) = |\langle\Psi^-|\phi_{st}(a, b)\rangle|^2.$$

For arbitrary directions $a, b \in S^2$ with relative angle $\theta = \arccos(a \cdot b)$, direct evaluation yields

$$P_{++} = P_{--} = \frac{1}{2} \sin^2\left(\frac{\theta}{2}\right), \quad P_{+-} = P_{-+} = \frac{1}{2} \cos^2\left(\frac{\theta}{2}\right).$$

The correlation function is therefore

$$E(a, b) = \sum_{s, t \in \{+, -\}} st P_{st} = -\cos \theta = -a \cdot b.$$

3.3 Geometric visualisation

Fubini-Study Voronoi geometry provides a useful projective exemplar for context-dependent outcome regions Blore [2026d,e]. In the present singlet case, the symmetric configuration $\theta = \pi/2$ admits a simple visualisation: the singlet ray is equidistant in the Fubini-Study metric from the four product outcome rays. We do not use this as a derivation route. In general, Voronoi-cell geometry does not by itself determine the Born values for sharply prepared pure states, since the probabilities are state-conditioned rather than state-independent volume fractions.

3.4 Continuity and rotational covariance

The singlet state is invariant under the diagonal $SU(2)$ action:

$$(U \otimes U)|\Psi^-\rangle = e^{i\phi(U)}|\Psi^-\rangle.$$

Accordingly, simultaneous rotation of the detector directions induces the corresponding transformation of the joint outcome projectors while leaving the singlet preparation invariant up to phase. The resulting correlation function therefore depends only on the rotational scalar $a \cdot b$, so

$$E(a, b) = f(a \cdot b)$$

for some continuous function f .

Continuity follows from the smooth dependence of the outcome regions on the detector settings together with the absolute continuity of the epistemic preparation measure, as recorded in Appendix A. Since the explicit Born-form evaluation in Section 3.2 yields

$$E(a, b) = -\cos \theta = -a \cdot b,$$

the admissible function is fixed in the singlet case.

4 CHSH consequence and programme placement

For the standard coplanar choice of settings, substitution of

$$E(a, b) = -a \cdot b$$

into the CHSH expression yields

$$|S| = 2\sqrt{2}.$$

Thus the singlet specialisation considered here reproduces the standard Bell-violating value.

The Bell-facing structural diagnosis is not developed in the present note. In the current programme architecture, that role belongs to c1, which formalises the Fine obstruction for context-indexed outcome maps and clarifies why Bell violation excludes a single measurable global assignment reproducing all incompatible settings simultaneously. The constructive system-apparatus Hamiltonian model is likewise deferred to TN3.

5 Scope and conclusion

The present note establishes only the finite-dimensional singlet specialisation of the symmetry-and-operations route. It does not provide a constructive measurement Hamiltonian, a basin-dynamics derivation of the correlation function, or the primary Bell-premise diagnosis. Those tasks are carried by TN3 and c1 respectively. Nor does the present note attempt to restate the full finite-dimensional closure package. Its role is narrower: to show that the two-qubit singlet in dimension 4 lies within the existing measure-and-weight architecture.

Operational no-signalling for singlet marginals is handled elsewhere in the finite-dimensional stack. At the structural Bell level, c1 proves no-signalling for context-indexed outcome maps under local implementation assumptions. At the mixed-state and subsystem level, TN4 establishes marginal stability and reduced-state consistency.

For the two-qubit singlet in $H_{AB} \cong \mathbb{C}^4$, the finite-dimensional symmetry-and-operations package yields the Born-form assignment, and direct evaluation gives

$$E(a, b) = -a \cdot b.$$

The standard CHSH value $2\sqrt{2}$ follows.

A Continuity of Volume Under Smooth Partition Deformation

A.1 Purpose

Appendix B invoked continuity of the correlation function with respect to the detector angle θ . The present appendix justifies this claim at the level of measure theory. We show that if outcome regions deform smoothly under variation of the measurement settings, and the preparation measure is absolutely continuous, then the induced probabilities vary continuously.

A.2 Set-up

Let $M = CP^3$ be equipped with the Fubini-Study measure μ_{FS} , and let $\rho_{\text{ep}, \Psi^-}([\psi])$ denote the preparation density of the singlet ensemble, assumed absolutely continuous with respect to μ_{FS} :

$$d\nu_{\Psi^-}^{\text{ep}} = \rho_{\text{ep}, \Psi^-} d\mu_{FS}, \quad \rho_{\text{ep}, \Psi^-} \in L^1(M), \quad \rho_{\text{ep}, \Psi^-} \geq 0.$$

For each detector setting parameter $\theta \in [0, \pi]$, let

$$\Omega_{st}(\theta) \subset M$$

denote the measurable outcome region corresponding to (s, t) .

Assume:

- (A1) The boundaries $\partial\Omega_{st}(\theta)$ are piecewise smooth hypersurfaces in M .
- (A2) The dependence $\theta \mapsto \partial\Omega_{st}(\theta)$ is C^1 in the Hausdorff topology.
- (A3) $\mu(\partial\Omega_{st}(\theta)) = 0$ for all θ .

Condition (A3) holds because smooth hypersurfaces in a Riemannian manifold have codimension one and hence μ_{FS} -measure zero.

A.3 Continuity of the Probability Functional

The operational probability for outcome (s, t) is

$$P_{st}(\theta) = \int_{\Omega_{st}(\theta)} \rho_{\text{ep}, \Psi^-}([\psi]) d\mu_{FS}([\psi]).$$

Let $\theta_n \rightarrow \theta$. Under assumptions (A1)–(A3), the symmetric difference

$$\Omega_{st}(\theta_n) \triangle \Omega_{st}(\theta)$$

has measure tending to zero as $n \rightarrow \infty$. More precisely, under the C^1 deformation assumption on the boundary family, each boundary $\partial\Omega_{st}(\theta)$ admits a tubular neighbourhood of arbitrarily small volume. For θ_n sufficiently close to θ , the symmetric difference is contained in such a tubular neighbourhood, so its μ_{FS} -measure tends to zero.

Since $\rho_{\text{ep}, \Psi^-} \in L^1(M)$ and is non-negative, the dominated convergence theorem implies

$$\lim_{n \rightarrow \infty} P_{st}(\theta_n) = P_{st}(\theta).$$

Thus $P_{st}(\theta)$ is continuous in θ .

A.4 Continuity of the Correlation Function

The correlation function is defined by

$$E(\theta) = \sum_{s,t \in \{+, -\}} st P_{st}(\theta).$$

Since each $P_{st}(\theta)$ is continuous, $E(\theta)$ is continuous on $[0, \pi]$.

A.5 Remarks

1. The proof does not assume any specific functional form of the probabilities.
2. The result depends only on absolute continuity of the preparation measure and smooth deformation of the partition boundaries.
3. No stochastic hypothesis enters; continuity is a purely geometric consequence of smooth partition motion on the compact manifold M .

This establishes the continuity assumption used in Appendix B and Section 3.

B Verification of the Symmetry-and-Operations Hypotheses in Dimension Four

B.1 Purpose

Section 2.5 invokes the finite-dimensional measure-and-weight layer supplied by Paper B together with the measure-bridge support note TN1 Blore [2026c,f]. The present appendix verifies the singlet-specialised operational package used for that application in the two-qubit case.

We restrict attention to rank-one projectors on $\mathcal{H}_{AB} \cong \mathbb{C}^4$.

B.2 Singlet-specialised hypothesis package

In the present note we do not restate the full general theorem of the finite-dimensional probability layer in maximal generality. Rather, we verify the singlet-specialised operational package needed for the application in $H_{AB} \cong \mathbb{C}^4$:

- (H1) non-negativity,
- (H2) normalisation and finite additivity on each orthogonal projective resolution,
- (H3) continuity under smooth variation of the measurement setting,
- (H4) covariance under the residual diagonal $SU(2)$ symmetry of the singlet setting,
- (H5) operational projector consistency within a fixed measurement context.

These are the singlet-level conditions used in Section 2.5 together with the existing finite-dimensional measure-and-weight layer.

We note that the singlet-specific role of H4 in the present note is narrower than the role of the general symmetry assumptions in the finite-dimensional measure-and-weight layer. The latter are supplied by Paper B together with the measure bridge of TN1. Here, H4 records the residual diagonal $SU(2)$ symmetry of the singlet preparation, which is used in §3.4 to constrain the correlator to depend on $a \cdot b$.

B.3 Definition of the Induced Functional

In the present construction, the probability functional is defined by

$$p_{\text{prep}}(\Pi_{st}(a, b)) = \mu_{\text{prep}}(\Omega_{st}(a, b)),$$

where μ_{prep} is the pushforward of the Liouville measure restricted to the preparation ensemble.

We verify each hypothesis.

B.4 (H1) Non–Negativity

Since μ_{prep} is a positive measure and $\Omega_{st}(a, b)$ is measurable,

$$p_{\text{prep}}(\Pi_{st}(a, b)) = \mu_{\text{prep}}(\Omega_{st}(a, b)) \geq 0.$$

B.5 (H2) Additivity on Orthogonal Resolutions

For fixed (a, b) , the projectors $\{\Pi_{st}(a, b)\}$ form an orthogonal resolution of the identity.

The corresponding regions $\{\Omega_{st}(a, b)\}$ form a measurable partition of the preparation support (up to null sets). Hence,

$$\sum_{s,t} p_{\text{prep}}(\Pi_{st}(a, b)) = \sum_{s,t} \mu_{\text{prep}}(\Omega_{st}(a, b)) = 1.$$

B.6 (H3) Continuity

Appendix A established that if the apparatus–induced partitions depend smoothly on detector settings, then

$$(a, b) \mapsto p_{\text{prep}}(\Pi_{st}(a, b))$$

is continuous.

Since rank–one projectors vary smoothly over $\mathbb{CP}^3$, the functional $p_{\text{prep}}(\Pi)$ is continuous.

B.7 (H4) Residual symmetry covariance

The singlet state is invariant under the diagonal action of $SU(2)$:

$$(U \otimes U)|\Psi^-\rangle = e^{i\phi(U)}|\Psi^-\rangle.$$

Accordingly, simultaneous rotation of the detector directions induces the corresponding transformation of the joint rank–one outcome projectors while leaving the singlet preparation invariant up to phase. The induced probability assignment is therefore covariant under the residual diagonal $SU(2)$ symmetry relevant to the singlet setting.

This is the covariance used in Section 3.4 to conclude that the correlator depends only on $a \cdot b$.

B.8 (H5) Operational Consistency

If a rank–one projector Π corresponds to a fixed macroscopic outcome channel realised in different compatible orthogonal resolutions, then the associated ontic basin is the same measurable region (up to null sets). Therefore

$$p_{\text{prep}}(\Pi)$$

is independent of the choice of compatible orthogonal completion.

This ensures operational projector consistency without assuming noncontextual value assignments. H5 is the structural condition that plays the role of Gleason-type noncontextuality in the finite-dimensional selection argument. Within the present non-constructive note, it is recorded as a hypothesis on the relation between apparatus couplings and outcome projectors, rather than derived from the ontic dynamics. An explicit dynamical verification, exhibiting system–apparatus Hamiltonians for which the basin attached to a pointer state is independent of the orthogonal completion, is deferred to TN3.

B.9 Conclusion

The induced probability functional satisfies hypotheses (H1)–(H5). Therefore, by the symmetry-and-operations uniqueness theorem in dimension four, there exists a density operator ρ_{prep} such that

$$p_{\text{prep}}(\Pi) = \text{Tr}(\rho_{\text{prep}}\Pi).$$

In particular, for the singlet preparation,

$$\rho_{\text{prep}} = |\Psi^-\rangle\langle\Psi^-|,$$

and the Born rule follows as a structural consequence of deterministic volume geometry.

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